SURVEYING & LEVELLING · NOTE SLO5

Compass Survey — Meridians, Bearings, Prismatic Compass and Local Attraction

Course topics: SWCE1.4 L6

Direction of a line

The direction of a line is the horizontal angle it makes with a fixed reference line, the **meridian**.

Meridian	Defined by	Behaviour and use
True meridian	True north–south	Does not change; used in geodetic surveys
Magnetic meridian	A freely suspended magnetic needle	Subject to variation; used in plane surveys
Assumed meridian	Any arbitrarily chosen direction	Small local surveys

- **Magnetic declination** is the angle between the true and magnetic meridians. It is **east** when the needle points east of true north and **west** when it points west. Obtain it from the nearest observatory.
- **Azimuth** — clockwise horizontal angle from north, 0° to 360°.
- **Bearing** — angle a line makes with the magnetic meridian, measured with a compass.

Prismatic compass

Use it to find the bearing of a line with respect to the magnetic meridian.

- Circular box 6–15 cm in diameter, with a magnetic needle pivoted at the centre.
- A graduated aluminium ring (0–360°) moves with the needle; graduations run clockwise with zero at the **south end** of the needle, so the reading under the prism is zero when the needle points north and bearings read clockwise from north.
- A sight vane with a hair line, a prism to magnify the graduations, a glass cover, a mirror and dark glasses.
- Mounted on a tripod with a ball-and-socket joint.

Taking a bearing

1. Centre the compass over the station.
2. Level it so the ring swings freely.
3. Bisect the object with the hair line of the sight vane.
4. Let the needle settle and read the bearing through the prism.

Included angle between two lines

- Take the bearings of both lines; the difference is the angle.
- If the difference exceeds 180°, the interior angle = 360° - difference.

Fore and back bearings

- **Fore bearing (FB):** bearing in the direction of survey. **Back bearing (BB):** bearing in the reverse direction.

- Add 180° when FB is below 180°; subtract 180° when FB is above 180°.

$$\text{BB} = \text{FB} \pm 180°$$

> FB of AB = 60° → BB = 60° + 180° = 240°. FB of BC = 250° → BB = 250° - 180° = 70°.

Local attraction

- **Cause:** nearby magnetic objects — iron, steel, electric wires, iron ore — deflect the needle.
- **Detection:** if the FB and BB of a line differ by exactly 180°, there is no local attraction; otherwise it exists.
- **Key property:** local attraction at a station shifts every bearing read there by the same amount, so the **included angles stay correct**.
- **Correction:** start from a line free of local attraction, use it as the reference and compute the correct bearings of the following lines.

Prismatic compass traverse

- **Field steps:** reconnaissance, marking stations, running survey lines, taking bearings, measuring chain lengths, taking offsets and booking field notes.
- **Choose stations** that are intervisible, easy to chain between and near the boundaries; prefer long lines.

1. Centre and level the compass at a station.
2. Take the FB of the forward line and the BB of the previous line.
3. Chain the line and repeat at every station; plot offsets and details.

Intersection method: for an inaccessible point, take its bearings from two stations; the rays intersect at the point.

Plotting and closing error

- Plot by the **parallel meridian method:** fix the first station, draw the magnetic meridian, lay off the first line by bearing and length, draw a parallel meridian through the next station and repeat around the traverse.
- **Closing error** — failure of a closed traverse to meet its starting point; caused by chaining and bearing errors.
- Re-survey if the error is large; if small, distribute it in proportion to the lengths of the sides.

Errors in compass work

Error	Remedy
Bent needle	Read both ends and take the mean
Bent pivot	Adjust the pivot until the two readings differ by 180°
Plane of sight not vertical	Test with a plumb bob
Sluggish needle	Tap lightly or re-magnetise
Reading error	Keep the eye vertically above the graduation
Undetected magnetic variation	Keep iron objects away

Sources: ICAR eCourse 'Surveying and Leveling' — course content by Dr D. Tamil Mani (Dept. of Soil & Water Conservation Engg., AEC&RI, TNAU Coimbatore), reviewed by Dr V. M. Abdul Hakkim (KCAET, Kerala Agricultural University); published as an AgriMoon e-book. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. The fore bearing of a line is 60°. Its back bearing is

(a) 120° (b) 180°
(c) 240° (d) 300°

Q2. Local attraction at a station is indicated when FB and BB of a line

(a) differ by exactly 180° (b) do not differ by exactly 180°
(c) are equal (d) differ by 90°

Q3. Local attraction does NOT affect

(a) bearings (b) included angles
(c) fore bearings (d) back bearings

Q4. The angle between true and magnetic meridians is

(a) dip (b) azimuth
(c) magnetic declination (d) bearing

Q5. A compass traverse is plotted by the

(a) radiation method (b) parallel meridian method
(c) intersection method (d) grid method

Answers

Q1 — (c) 240° — *BB = FB + 180° when FB < 180°.*
Q2 — (b) do not differ by exactly 180°
Q3 — (b) included angles — *All bearings at the station shift equally.*
Q4 — (c) magnetic declination
Q5 — (b) parallel meridian method